

① ex 4.2.1

Find extremal for functional $\rightarrow J(x) = \int_0^{\pi/2} \dot{x}^2(t) - x^2(t) dt$
 $x^* = ? = \text{extremal}$

B.C. $x(0) = 0$
 $x(\frac{\pi}{2}) = 1$

노 23~26 쪽 (t_0 와 t_f = 고정, $x(t_0)$ 와 $x(t_f)$ = 고정) The simplest variation problem.

교재 128 쪽

710x 1

$\frac{\partial \mathcal{H}}{\partial x}(x^*, \dot{x}^*, t) - \frac{d}{dt} \left[\frac{\partial \mathcal{H}}{\partial \dot{x}}(x^*, \dot{x}^*, t) \right] = 0 \rightarrow$ Necessary condition for x^* to be an extremal.

E-L eq when $x(t_0), x(t_f)$ are given. (243)

$\mathcal{H} = \dot{x}^2 - x^2$, $\frac{\partial \mathcal{H}}{\partial x} = -2x$, $\frac{\partial \mathcal{H}}{\partial \dot{x}} = 2\dot{x}$, $\frac{d}{dt}(2\dot{x}) = 2\ddot{x} \rightarrow$ EL eq에 대입 $\rightarrow -2x - 2\ddot{x} = 0$
 $x = -\ddot{x}$, $\ddot{x} = -x$

$\ddot{x} = -x$ 의 해는 $x(t) = A \cos t + B \sin t$ 이다. 경계조건으로, $x(0) = 0 = A + 0 \Rightarrow A = 0$, $x(\frac{\pi}{2}) = 1 = 0 + B \Rightarrow B = 1$
 $\therefore x(t) = \sin t$
 $\therefore x^*(t) = \sin t$
 extremal
 > 3중의 extremal
 x^* 이다.

② ex 4.2.2

Determine the smooth curve of smallest length connecting point $x(0)=1$ to the line $t=5$.

It can be shown that the length of a curve lying in the $t-x(t)$ plane, with $t_0=0$, $t_f=5$, is

$J(x) = \int_0^5 \sqrt{1+\dot{x}^2} dt //$

노 27 쪽
 교재 130 쪽

(t_0 와 $x(t_0)$ = 고정, t_f = 고정, $x(t_f)$ = 자유) 710x 2
 $J(x) = \int_0^5 \sqrt{1+\dot{x}^2} dt$, $\mathcal{H} = (1+\dot{x}^2)^{0.5}$
 $x(0) = 1$, $x(5) = \text{free}$.

① EL 식, $\frac{\partial \mathcal{H}}{\partial x} - \frac{d}{dt} \left(\frac{\partial \mathcal{H}}{\partial \dot{x}} \right) = 0$ / $\frac{\partial \mathcal{H}}{\partial x} = 0$, $\frac{\partial \mathcal{H}}{\partial \dot{x}} = \frac{2\dot{x}}{2\sqrt{1+\dot{x}^2}} = \frac{\dot{x}}{\sqrt{1+\dot{x}^2}}$
 $\frac{d}{dt} \left(\frac{\dot{x}}{\sqrt{1+\dot{x}^2}} \right) = \frac{\ddot{x}}{\sqrt{1+\dot{x}^2}} - \frac{\dot{x}^2 \cdot \ddot{x}}{(1+\dot{x}^2)^{1.5}} = 0$

② 횡단식.
 $\frac{\partial \mathcal{H}}{\partial \dot{x}} = 0$
 $t = t_f$

$\frac{\partial \mathcal{H}}{\partial \dot{x}} = \frac{2\dot{x}}{2\sqrt{1+\dot{x}^2}} = \frac{\dot{x}}{\sqrt{1+\dot{x}^2}} = \frac{\dot{x}(t_f)}{\sqrt{1+\dot{x}(t_f)^2}} = 0 = \frac{A}{\sqrt{1+A^2}} = 0$
 $\therefore A = 0$

$\Rightarrow \frac{\ddot{x}}{\sqrt{1+\dot{x}^2}} \left(1 - \frac{\dot{x}^2}{1+\dot{x}^2} \right) = 0$

$\ddot{x} = 0$ or $\dot{x}^2 = 1$
 $\therefore \dot{x} = \text{const} = A$
 $\therefore x = At + B$ (일차식) $\rightarrow x = At + 1$
 $x(0) = 1 \Rightarrow B = 1$

$\therefore x(t) = 1$
 $t_0 = 0$ $x(t_0) = 1$
 $t_f = 5$ $x(t_f) = 1$

③ ex 4.2.3. Determine extremal for functional $J(x) = \int_0^2 \dot{x}^2 + 2x\dot{x} + 4x^2 dt$. $x(0)=1$, $x(2)=2$.

노드 27쪽
교재 130쪽.

t_0 와 $x(t_0)$ 고정, t_f 고정, $x(t_f)$ 자유.

710K 2

$$g = \dot{x}^2 + 2x\dot{x} + 4x^2$$

① EL 1) $\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) = 0$, $\frac{\partial g}{\partial x} = 2\dot{x} + 8x$, $\frac{\partial g}{\partial \dot{x}} = 2\dot{x} + 2x$, $\frac{d}{dt}(2\dot{x} + 2x) = 2\ddot{x} + 2\dot{x} \Rightarrow 2\ddot{x} + 8x - 2\dot{x} - 2\dot{x} = 0$
 $4x = \ddot{x}$

② $t=t_f$

$\ddot{x} = 4x$ (2차 선형 방정식) $\rightarrow$ 해는 $x(t) = C_1 e^{2t} + C_2 e^{-2t}$ (고정된 초기 조건 4개). $\dot{x}(t) = 2C_1 e^{2t} - 2C_2 e^{-2t}$

$x(0) = C_1 + C_2 = 1$ (1개 경계조건).

③ 횡단식, $\frac{\partial g}{\partial \dot{x}} = 0$

$t=t_f$

$0 = 2\dot{x} + 2x \rightarrow x(t_f) = -\dot{x}(t_f)$, $\therefore x(2) = -\dot{x}(2)$

$x(2) = C_1 e^4 + C_2 e^{-4} \Leftrightarrow 2C_2 e^{-4} - 2C_1 e^4 = \dot{x}(2)$

$3C_1 e^4 = C_2 e^{-4}$, $C_2 = 3C_1 e^8$, $C_1 + C_2 = 1$

$\therefore x(t) = \frac{e^{2t}}{43e^8} + \frac{3e^8 \cdot e^{-2t}}{43e^8}$

$t_0 = 0, x(t_0) = 1$
 $t_f = 2, x(t_f) = \frac{e^4 + 3e^4}{43e^8} = \frac{4e^4}{43e^8} //$

$C_1 + C_2 e^8 = 1 \rightarrow C_1 (1 + 3e^8) = 1$

$C_1 = \frac{1}{1 + 3e^8}$, $C_2 = 1 - \frac{1}{1 + 3e^8} = \frac{3e^8}{1 + 3e^8}$

④ ex 4.2.4 Find extremal for functional $J(x) = \int_1^{t_f} \left[2x + \frac{1}{2} \dot{x}^2 \right] dt$

$t_0 = 1, x(t_0) = 4$
 $t_f = 7, x(t_f) = 4$

노드 29쪽.
교재 134쪽.

t_0 와 $x(t_0)$ 고정, t_f 자유, $x(t_f)$ 고정.

710K 3

$g = 2x + \frac{\dot{x}^2}{2}$

① EL 1) $\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) = 0 \rightarrow \frac{\partial g}{\partial x} = 2$, $\frac{\partial g}{\partial \dot{x}} = \dot{x}$, $\frac{d}{dt} \dot{x} = \ddot{x} \rightarrow 2 - \ddot{x} = 0 \rightarrow \ddot{x} = 2$, $\dot{x} = 2t + A$, $x = t^2 + At + B$

② $t=t_f$

③ 횡단식, $g - \frac{\partial g}{\partial \dot{x}} \cdot \dot{x} = 0 \rightarrow 2x + \frac{\dot{x}^2}{2} - \dot{x} \cdot \dot{x} = 2x - \frac{\dot{x}^2}{2} = 0 \rightarrow 4x = \dot{x}^2 \rightarrow \frac{4x(t_f)}{4} = \frac{\dot{x}(t_f)^2}{4} \rightarrow \dot{x}(t_f) = 4$

④ $t=t_f$

① $\dot{x}(t_f) = 4 \Rightarrow 2t_f + A = 4$
 ② $4A + B = 4 \Rightarrow A + B = 3$
 ③ $t_f^2 + At_f + B = 4$
 ④ $2t_f + A = 4$

$4(2t_f + A) = 4(t_f^2 + At_f + B) \Rightarrow 4(2t_f + A) = 4t_f^2 + 4At_f + 4B$
 $4B = A^2$
 $A + B = 3$
 $A = 2, B = 1$
 $A = -6, B = 9$

노드 31쪽: 710K 3 문제 미제 있음.
 $J(x) = \int_0^{t_f} (\dot{x}^2 + x) dt$ $t_0 = 0, x(t_0) = 0$
 $t_f = 2, x(t_f) = 1$

$2t_f + A = 4$
 $\therefore A = 2, t_f = 1$ (X)
 $\therefore A = -6, t_f = 5$ (O)

$\therefore x(t) = t^2 - 6t + 9$ $t = 1 \sim 5$
 $x(t) = 4 \sim 4$

5.3.4.2 문제 4.13
 $J(x) = \int_0^{t_f} \sqrt{1+\dot{x}^2} dt$

$x(0)=2$

$x(t)$ 는 $\theta(t) = 5-4t$ 에서 끝난다.

5.3.4 문제 4.13에 t_0 와 $x(t_0)$ = 고정

고정 문제 4.13. t_f 와 $x(t_f)$ = 자유

문제 4.13

① $\frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = 0$ $t=t$
 고정점.

$\mathcal{L} = \sqrt{1+\dot{x}^2} = (1+\dot{x}^2)^{0.5}$

$x(t_f) = \theta(t_f) = 5-4t_f$

$\frac{\partial \mathcal{L}}{\partial x} = 0$

$\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{2\dot{x}}{2\sqrt{1+\dot{x}^2}} = \frac{\dot{x}}{\sqrt{1+\dot{x}^2}}$

$\frac{d}{dt} \left(\frac{\dot{x}}{\sqrt{1+\dot{x}^2}} \right) = \frac{d}{dt} \left(\dot{x} \cdot (1+\dot{x}^2)^{-0.5} \right)$

$= \ddot{x} \cdot (1+\dot{x}^2)^{-0.5} + \dot{x} \cdot -0.5 \cdot (1+\dot{x}^2)^{-1.5} \cdot 2\dot{x}\ddot{x}$

$= \frac{\ddot{x}}{\sqrt{1+\dot{x}^2}} - \frac{\dot{x}^2 \ddot{x}}{(1+\dot{x}^2)^{1.5}} = 0$

$\Rightarrow \frac{\ddot{x}}{\sqrt{1+\dot{x}^2}} \left(1 - \frac{\dot{x}^2}{1+\dot{x}^2} \right) = 0$

$\forall t: 0 \leq t \leq t_f$

② $\mathcal{L} + \frac{\partial \mathcal{L}}{\partial \dot{x}} [\dot{\theta} - \dot{x}] = 0$ $t=t_f$
 고정점.

$\mathcal{L}(x(t_f), \dot{x}(t_f), t_f) + \frac{\partial \mathcal{L}}{\partial \dot{x}}(x(t_f), \dot{x}(t_f), t_f) [\dot{\theta}(t_f) - \dot{x}(t_f)] = 0$

$\frac{\dot{x}}{\sqrt{1+\dot{x}^2}} [-4 - \dot{x}] + \sqrt{1+\dot{x}^2} = 0$

$\frac{-4\dot{x} - \dot{x}^2}{\sqrt{1+\dot{x}^2}} + \frac{1+\dot{x}^2}{\sqrt{1+\dot{x}^2}} = 0 = \frac{1-4\dot{x}}{\sqrt{1+\dot{x}^2}} \Rightarrow \dot{x} = \frac{1}{4}$

$\therefore \dot{x}(t_f) = \frac{1}{4}$

①에서 얻은 조건에서 $x(t) = At + 2$ 를 미분하면 $\dot{x}(t) = A$

$\therefore A = \frac{1}{4}$ 이다!

$\therefore x(t) = \frac{t}{4} + 2$

이때, $x(t_f) = \theta(t_f)$ 이므로

$\frac{t_f}{4} + 2 = 5 - 4t_f$

$\therefore t_f = \frac{12}{17}$

정답!

$\therefore \ddot{x}(t) = 0$ ③

$\therefore \dot{x}(t) = A$ (상수)

$\therefore x(t) = At + B$ (일차함수)

$x(0) = 2 = B$

$x(t) = At + 2$

EX 4.2.5

Find

extremal for functional

$J(x) = \int_0^{t_f} \sqrt{1+\dot{x}^2} dt$

$t_0 = 0, x(t_0) = 0$

$t_f = x_f, x(t_f) = x_f$

$\theta(t) = 15-5t$

① $\frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = 0 \rightarrow \frac{\partial \mathcal{L}}{\partial x} = 0, \frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{2\dot{x}}{2\sqrt{1+\dot{x}^2}}$

$t=t$

$t=t$

$0 = \frac{\ddot{x}}{\sqrt{1+\dot{x}^2}} \left(1 - \frac{\dot{x}^2}{1+\dot{x}^2} \right) = \frac{\ddot{x}}{\sqrt{1+\dot{x}^2}} - \frac{\dot{x}^2 \ddot{x}}{(1+\dot{x}^2)^{1.5}} = 2\dot{x}\ddot{x}$

$\forall t: 0 \leq t \leq t_f$

$\therefore \ddot{x} = 0$
 $\dot{x} = \text{상수} = A$
 $x = At + B = \text{일차함수}$

$x(0) = 0 = B \therefore x = At$

② $\mathcal{L} + \frac{\partial \mathcal{L}}{\partial \dot{x}} (\dot{\theta} - \dot{x}) = 0$

$t=t_f$

$\sqrt{1+\dot{x}^2} + \frac{\dot{x}}{\sqrt{1+\dot{x}^2}} \cdot (-5 - \dot{x}) = 0 = \frac{1+\dot{x}^2 - 5\dot{x} - \dot{x}^2}{\sqrt{1+\dot{x}^2}} = \frac{1-5\dot{x}}{\sqrt{1+\dot{x}^2}} = 0$

$\dot{x} = \frac{1}{5} = A$

$\therefore x = \frac{t}{5}$

$x(t_f) = \frac{t_f}{5} = 15-5t_f = \theta(t_f)$

$\therefore t_f = 2.885$

$x(t_f) = \frac{t_f}{5} = 0.577$

ex 4.2.6

$$\frac{\partial \theta}{\partial x} - \frac{1}{A} \left(\frac{\partial \theta}{\partial \dot{x}} \right) = 0$$

(1) ELA) $t=t_0$

$$J(x) = \int_{t_0}^{t_f} \sqrt{1 + \dot{x}^2} dt$$

$$t_0 = 0, x(t_0) = 0$$

$$\theta(t) = \frac{1}{2} (t-5)^2 - \frac{1}{2}$$

$$7110 \leq 4$$

(4)

ex 4.2.5)의 ELA 결과로부터 $\ddot{x} = 0, \dot{x} = A, x = At + \frac{B}{0} = At$ 이다.

$$\textcircled{!!} \text{필터} \textcircled{!} t=t_f \quad \theta + \frac{\partial \theta}{\partial \dot{x}} [\dot{\theta} - \dot{x}] = 0 \rightarrow \frac{\sqrt{1 + \dot{x}^2}}{\sqrt{1 + \dot{x}^2}} + \frac{\dot{x}}{\sqrt{1 + \dot{x}^2}} \cdot (t-5 - \dot{x}) = 0 \rightarrow \frac{1 + \dot{x}^2 + t\dot{x} - 5\dot{x} - \dot{x}^2}{\sqrt{1 + \dot{x}^2}} = 0$$

$$1 + (t-5)\dot{x} = 0 \rightarrow t-5 = -\frac{1}{A}$$

$$\therefore x(t) = \frac{1}{2} \quad t=0 \sim 3, x(t)=0 \sim 1.5$$

$$x(t) = \theta(t) = At_5 = \frac{1}{2} (t_5 - 5)^2 - \frac{1}{2}$$

$$2At = \frac{1}{A^2} - 1$$

$$A=0.5, t_5=3$$

예제 보드 기종

교재 134쪽

t_f 만 248 $x(t_0) \cdot t_0 = \text{고정}$

$$7110 \leq 3$$

$$J(x) = \int_0^{t_f} (\dot{x}^2 + x) dt$$

$$x(0) = 0$$

$$x(t_f) = 1$$

$$t_f = 248$$

$$\theta = \dot{x}^2 + x$$

$$\ddot{x} = \frac{1}{2}$$

(1) ELA) $t=t$

$$\frac{\partial \theta}{\partial x} - \frac{1}{A} \left(\frac{\partial \theta}{\partial \dot{x}} \right) = 0 \rightarrow -2\dot{x} + 1 = 0, x = At^2 + Bt + C \text{ (이차식)}$$

$$\rightarrow x(0) = 0, C = 0, \rightarrow x(t_f) = At_f^2 + Bt_f = 1$$

$$\rightarrow \dot{x} = 2A \rightarrow A = \frac{1}{4}$$

(!!) 필터 $t=t_f$

$$\theta - \frac{\partial \theta}{\partial \dot{x}} \cdot \dot{x} = 0 \rightarrow \dot{x}^2 + x - 2\dot{x}^2 = 0 \rightarrow x - \dot{x}^2 = 0 \rightarrow \dot{x} = 2At + B = \frac{1}{2}t + B$$

$$\rightarrow \dot{x}^2 = \frac{1}{4}t^2 + Bt + B^2$$

$$x = \dot{x}^2 \Rightarrow \frac{1}{4}t_f^2 + Bt_f = \frac{1}{4}t_f^2 + Bt_f + B^2$$

$$t=t_f$$

$$B^2 = 0 = B$$

$$\therefore x(t) = \frac{t^2}{4}, x(t_f) = 1 = \frac{t_f^2}{4} \rightarrow t_f = 2 //$$